

A certification criterion for Maass forms on $\Gamma_0(1951)$ with nebentypus χ_{780}

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Abstract

We give a version of the Booker–Strömbergsson–Venkatesh criterion [1] for the group $\Gamma_0(N)$ with a primitive character, and make it explicit for $N = 1951$, $\chi = \chi_{780}$. Given any *odd* finite Fourier series $\tilde{f}$ whose terms are eigenfunctions of Δ with eigenvalue $1/4$, the criterion bounds the distance from $1/4$ to the spectrum of the odd cusp forms by the L^2 norms of the jump of an explicit automorphization of $\tilde{f}$ across the 921 sides of a fundamental domain, and of the jump of its normal derivative. Three changes from [1] make it practical at this level: oddness removes the continuous spectrum, so no Hecke operator is needed; Green’s identity replaces the supremum over a two-dimensional neighbourhood of the boundary by one-dimensional integrals over the sides; and the Fricke involution enters only through the definition of the automorphization. The fundamental domain has been computed exactly and checked against the volume, the class numbers and the genus.

1 Setting

Let $N = 1951$ (prime) and let $\chi = \chi_{780}$ be the Dirichlet character modulo N with Conrey label 780: it is primitive, of order 5, and even. Write $\Gamma = \Gamma_0(N)$, $W = \frac{1}{\sqrt{N}} \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$, $\Gamma^+ = \langle \Gamma, W \rangle$, and $\iota(z) = -\bar{z}$. A function F on $\mathbb{H}$ is χ -*automorphic* if $F(\gamma z) = \chi(d_\gamma)F(z)$ for $\gamma = \begin{pmatrix} a_\gamma & b_\gamma \\ c_\gamma & d_\gamma \end{pmatrix} \in \Gamma$. Let $\Delta = -y^2(\partial_x^2 + \partial_y^2)$ and let $L_-^2(\Gamma \backslash \mathbb{H}, \chi)$ be the odd ($F \circ \iota = -F$) χ -automorphic functions with $\int_{\Gamma \backslash \mathbb{H}} |F|^2 d\mu < \infty$, $d\mu = dx dy/y^2$.

The approximation is a finite, exactly odd Fourier series

$$\tilde{f}(z) = \sum_{0 < |n| \leq M} c_n \sqrt{y} K_0(2\pi|n|y) e(nx), \quad c_{-n} = -c_n, \quad (1)$$

and $\tilde{f}^*(z) := \overline{\tilde{f}(\iota z)} = \sum \bar{c}_n \sqrt{y} K_0(2\pi|n|y) e(nx)$. Every term is an eigenfunction of Δ with eigenvalue exactly $1/4$, so $\tilde{f}$ and $\tilde{f}^*$ are too, on all of $\mathbb{H}$. The number $\tilde{\lambda} \in \mathbb{C}^\times$ is arbitrary; in practice it is the float64 value of the Fricke eigenvalue $\tau(\chi)\zeta_5^3/\sqrt{N}$. Nothing below requires $|\tilde{\lambda}| = 1$ or $\tilde{f}$ to satisfy any relation exactly: the relations enter only through the size of the jumps.

2 The fundamental domain

Every element of Γ^+ with $c \neq 0$ has isometric circle $|cz + d| = 1$ (determinant normalised to one). Writing the centre as a reduced fraction u/v , the circle is unique and

$$\text{radius}^2 = \begin{cases} 1/(v^2 N) & N \nmid v \quad (\text{coset } W\Gamma), \\ 1/v^2 & N \mid v \quad (\Gamma). \end{cases}$$

Let $\mathfrak{F}^+ = \{| \operatorname{Re} z| \leq \frac{1}{2}\}$ minus the open disks bounded by these circles: the Ford domain of Γ^+ . It is convex. Two semicircles centred on $\mathbb{R}$ meet at most once in $\mathbb{H}$, so the floor of $\mathfrak{F}^+$ is computable exactly by inserting circles into an upper envelope; every intersection abscissa is rational.

Proposition 1 (computed exactly). *$\mathfrak{F}^+$ has 921 non-vertical sides (641 on circles of the coset $W\Gamma$, with denominators $v \leq 47$, and 280 on circles of Γ with $v = N$) and 922 vertices. Every side is paired with a side by an element of Γ^+ (65 sides with themselves). The vertices fall into 299 cycles: 297 accidental, one of order 2, one of order 3. Moreover:*

1. $\operatorname{vol}(\mathfrak{F}^+) = \frac{\pi}{6}(N+1) = 1022.0648\dots$, the value for $[\operatorname{PSL}_2(\mathbb{Z}) : \Gamma_0(N)]/2$;
2. the 66 elliptic points of order two (65 side midpoints and one vertex cycle) match $h(-4N) + h(-N) = 33 + 33$, the number of fixed points of W on $X_0(N)$;
3. Gauss–Bonnet then gives genus 65, which is $g(X_0(1951)^+) = (g(X_0(1951)) + 1)/2 - 66/4$;
4. the lowest vertex has height exactly $\sqrt{3}/(2N)$ (at $x = -153/(2N)$); the smallest interior angle is $\pi/3$; at most 6 tiles meet at a vertex;
5. the total hyperbolic length of the non-vertical sides is $L = 1290.5506\dots$; the longest side is the Fricke arc $|z| = 1/\sqrt{N}$, of length 8.9878, paired with itself by W .

The side pairings and vertex cycles are computed with exact integer matrices (an element of $W\Gamma$ is stored as $M/\sqrt{N}$ with M integral of determinant N); the angle sum around each of the 922 vertices equals 2π to forty digits. $\mathfrak{F} := \mathfrak{F}^+ \cup W\mathfrak{F}^+$ is a fundamental domain for Γ .

3 The automorphization

Define F on $\mathfrak{F}$ by

$$F = \tilde{f} \text{ on } \mathfrak{F}^+, \quad F(Wu) = \tilde{\lambda} \tilde{f}^*(u) \quad (u \in \mathfrak{F}^+),$$

and extend it to $\mathbb{H}$ by χ -automorphy. Then F is defined almost everywhere, χ -automorphic, odd (because $\mathfrak{F}^+ = \iota\mathfrak{F}^+$, $W\iota = \iota W$, $\iota\Gamma\iota = \Gamma$ with $d_{\iota\gamma\iota} = d_\gamma$, and $\tilde{f}, \tilde{f}^*$ are odd), and square integrable (it decays exponentially in both cusps, having no constant terms). On each tile $\tau = g^{-1}\mathfrak{F}^+$, $g \in \Gamma^+$, F coincides with a function G_τ that is an eigenfunction of Δ with eigenvalue $1/4$ on all of $\mathbb{H}$: for $g \in \Gamma$, $G_\tau(w) = \chi(a_g)\tilde{f}(gw)$; for $g \in W\Gamma$, $G_\tau(w) = \chi(a_{Wg})\tilde{\lambda}\tilde{f}^*(gw)$.

For a non-vertical side s of $\mathfrak{F}^+$ with pairing element g_s , the tile across s is $\tau_s = g_s^{-1}\mathfrak{F}^+$, and the jump of F across s is $-D_s$ with

$$D_s := G_{\tau_s} - \tilde{f}.$$

Across the vertical sides there is no jump ($\tilde{f}$ is 1-periodic). The jumps across all other edges of the tiling by Γ^+ are, up to unimodular factors and a factor $|\tilde{\lambda}|$, jumps across sides of $\mathfrak{F}^+$.

4 The criterion

Fix $\delta > 0$, put $Y = \sinh^{-2}(\delta/2)$ and $\phi(x) = 3(1-x)^2$ for $0 \leq x \leq 1$, $\phi(x) = 0$ for $x > 1$. The point-pair invariant $k(z, w) = \phi(Yu(z, w))$, $u = |z - w|^2/(4\operatorname{Im} z \operatorname{Im} w) = \sinh^2(d/2)$, is supported in $d(z, w) \leq \delta$, belongs to $C^{1,1}$, and k and ∇k vanish on the boundary of its support. Let $\Sigma = \bigcup_{g \in \Gamma^+} g(\partial\mathfrak{F}^+ \setminus \text{vertical sides})$ be the set of seams, and

$$\ell_{\max}(\delta) = \sup_{z \in \mathbb{H}} \operatorname{length}(\Sigma \cap B(z, \delta)).$$

Theorem 2. *Let*

$$Q_0 = \sum_s \int_s |D_s|^2 ds, \quad Q_1 = \sum_s \int_s |\partial_\nu D_s|^2 ds,$$

summed over the 921 non-vertical sides of $\mathfrak{F}^+$, with ds hyperbolic arc length and ∂_ν the hyperbolic unit normal derivative. Put

$$\begin{aligned} \kappa_1 &= 4\pi \int_0^1 |\phi'(x)| \sqrt{\frac{x}{Y}(1 + \frac{x}{Y})} dx, & G_1 &= \sup_{0 \leq x \leq 1} Y |\phi'(x)| \sqrt{\frac{x}{Y}(1 + \frac{x}{Y})}, \\ \alpha_0 &= \sqrt{\kappa_1 G_1 \ell_{\max}}, & \alpha_1 &= \sqrt{12\pi \ell_{\max}/Y}, & k_0 &= \frac{4\pi/Y}{\sqrt{1 + 1/Y}}, \\ \|\tilde{f}\|_{Y_1}^2 &= \sum_{0 < |n| \leq M} |c_n|^2 \int_{|n|Y_1}^\infty K_0(2\pi t)^2 \frac{dt}{t}, & Y_1 &= \frac{e^\delta}{\sqrt{N}}. \end{aligned}$$

If $\tilde{f} \neq 0$, there is an odd Maass cusp form for $\Gamma_0(N)$ with nebentypus χ whose Laplace eigenvalue λ_1 satisfies

$$\left| \lambda_1 - \frac{1}{4} \right| \leq \frac{\sqrt{1 + \max(1, |\tilde{\lambda}|^2)} (\alpha_0 \sqrt{Q_0} + \alpha_1 \sqrt{Q_1})}{k_0 \|\tilde{f}\|_{Y_1}}.$$

It is a newform, and may be taken to be a Hecke eigenform.

The proof occupies the next section. The quantities Q_0, Q_1 are the only inputs that depend on the computed coefficients beyond $\|\tilde{f}\|_{Y_1}$; they are one-dimensional integrals over the sides, and both points involved in each jump lie on the floor of $\mathfrak{F}^+$, at heights $\geq \sqrt{3}/(2N)$, which is above the sampling height of the Hejhal solve. No evaluation below the domain is needed.

5 Proof

Lemma 3 (no continuous spectrum). *$L_-^2(\Gamma \backslash \mathbb{H}, \chi)$ is contained in the cuspidal subspace, on which Δ has pure point spectrum.*

Proof. The continuous and residual spectrum are spanned by Eisenstein series $E_a(z, s)$ at the singular cusps and their residues. For the cusp ∞ , $E_\infty(z, s) = \sum_{\Gamma_\infty \backslash \Gamma} \bar{\chi}(d_\gamma) (\text{Im } \gamma z)^s$ and, since $\gamma \iota = \iota(\gamma \iota)$, $\text{Im } \iota w = \text{Im } w$ and $d_{\iota \gamma \iota} = d_\gamma$, one has $E_\infty(\iota z, s) = E_\infty(z, s)$; the cusp $0 = W\infty$ is the same with W , which commutes with ι . So every Eisenstein series, every analytic continuation and every residue is even, and odd functions are orthogonal to all of them. $\square$

Lemma 4 (jump representation). *Let $F_S = k * F$, i.e. $F_S(z) = \int_{\mathbb{H}} k(z, w) F(w) d\mu(w)$. Then F_S is smooth, odd, χ -automorphic, square integrable, and for every $z \in \mathbb{H}$*

$$(\Delta - \tfrac{1}{4})F_S(z) = \int_{\Sigma} \left(-[F](\sigma) \partial_\nu k(z, \sigma) + k(z, \sigma) [\partial_\nu F](\sigma) \right) ds(\sigma),$$

where $[\cdot]$ is the jump across the seam (in the direction opposite to ν).

Proof. Since k is a point-pair invariant, $\Delta_z k = \Delta_w k$, so $(\Delta - \frac{1}{4})F_S(z) = \sum_\tau \int_\tau (\Delta_w - \frac{1}{4})k(z, w) G_\tau(w) d\mu(w)$, a finite sum over the tiles meeting $B(z, \delta)$. On each tile apply Green's second identity to $k(z, \cdot)$ and G_τ . In hyperbolic terms it reads $\int_\tau (Ag - Bf) d\mu = -\oint_{\partial\tau} (g \partial_\nu f - f \partial_\nu g) ds$ with $\Delta f = A$, $\Delta g = B$ and outward ν , because $d\mu = dx dy / y^2$, $ds = |dz|/y$ and $\partial_\nu = y \partial_n$. Here $(\Delta - \frac{1}{4})G_\tau = 0$, and $k, \nabla k$ vanish on the edge of the support, so only $\partial\tau$ contributes. Each seam is shared by two tiles with opposite normals and adds the two one-sided terms, which gives the jumps. The vertical edges contribute nothing, since F and $\partial_\nu F$ are continuous across them. $\square$

Lemma 5 (Schur bound).

$$\|(\Delta - \tfrac{1}{4})F_S\|_{L^2(\Gamma \backslash \mathbb{H})} \leq \sqrt{1 + \max(1, |\tilde{\lambda}|^2)} (\alpha_0 \sqrt{Q_0} + \alpha_1 \sqrt{Q_1}).$$

Proof. Write the two terms of Lemma 4 as $T_j J_j$ with kernels $K_0 = \partial_\nu k$, $K_1 = k$ and densities $J_0 = [F]$, $J_1 = [\partial_\nu F]$. By Cauchy–Schwarz, $|T_j J_j(z)|^2 \leq (\int_\Sigma |K_j(z, \sigma)| ds) (\int_\Sigma |K_j(z, \sigma)| |J_j|^2 ds)$. The first factor is at most $\sup |K_j| \cdot \ell_{\max}$. Integrate over $\mathfrak{F}$ and unfold the second factor over Γ : $|J_j|$ is Γ -invariant and $|K_j(\gamma z, \gamma \sigma)| = |K_j(z, \sigma)|$, so $\int_{\mathfrak{F}} \int_\Sigma |K_j| |J_j|^2 \leq (\sup_\sigma \int_{\mathbb{H}} |K_j(z, \sigma)| d\mu(z)) \|J_j\|_{L^2(\Sigma_{\mathfrak{F}})}^2$, with $\Sigma_{\mathfrak{F}}$ any set meeting every Γ -orbit of seams. Take $\Sigma_{\mathfrak{F}} = \{\text{sides of } \mathfrak{F}^+\} \cup W\{\text{sides of } \mathfrak{F}^+\}$. For a side s paired by an element of $W\Gamma$, the edge $W(s)$ is Γ -equivalent to its partner s' ; for $g_s \in \Gamma$ a direct computation gives $|[F]|(Ww) = |\tilde{\lambda}| |D_{\iota(s)}(\iota w)|$, and likewise for normal derivatives. Hence $\|J_0\|_{L^2(\Sigma_{\mathfrak{F}})}^2 \leq (1 + \max(1, |\tilde{\lambda}|^2)) Q_0$, and the same for J_1, Q_1 .

For the kernels, $u = \sinh^2(d/2)$ has hyperbolic gradient of norm $\sqrt{u(1+u)}$, so $|\partial_\nu k| \leq Y|\phi'(Yu)|\sqrt{u(1+u)}$, whose supremum is G_1 and whose integral over $\mathbb{H}$ is at most κ_1 (the area element is $4\pi du$ after integrating over angles). For $K_1 = k$ the supremum is $\phi(0) = 3$ and the integral is $4\pi/Y$. This gives α_0 and α_1 . $\square$

Lemma 6 (the smoothing constant). *For an eigenfunction g of Δ with eigenvalue $1/4$ on a ball $B(z, \delta)$, $(k * g)(z) = \hat{k} g(z)$ with $\hat{k} \geq k_0 = \frac{4\pi/Y}{\sqrt{1+1/Y}}$.*

Proof. $\hat{k} = 4\pi \int_0^{1/Y} \phi(Yu) \omega(u) du$ with $\omega(u) = P_{-1/2}(1+2u)$ the spherical function. By $P_{-1/2}(\cosh d) = \frac{2}{\pi} \operatorname{sech}(d/2) K(\tanh(d/2))$ and $K \geq \pi/2$ one has $\omega(u) \geq (1+u)^{-1/2} \geq (1+1/Y)^{-1/2}$ on the support, and $\int \phi = 1$. $\square$

Lemma 7 (lower bound). $\|F_S\|_{L^2(\Gamma \backslash \mathbb{H})} \geq k_0 \|\tilde{f}\|_{Y_1}$.

Proof. Every isometric circle has radius at most $1/\sqrt{N}$, so the strip $\{|x| \leq \frac{1}{2}, y > 1/\sqrt{N}\}$ lies in $\mathfrak{F}^+$, and $F = \tilde{f}$ on its translates. If $\operatorname{Im} z \geq Y_1 = e^\delta/\sqrt{N}$ then $B(z, \delta)$ lies there, so $F_S(z) = \hat{k} \tilde{f}(z)$ by Lemma 6. Restrict the norm to $\{|x| \leq \frac{1}{2}, y \geq Y_1\} \subset \mathfrak{F}$ and use Parseval in x . $\square$

Proof of Theorem 2. $F_S \in L^2_-$ is in the domain of Δ , and by Lemma 3 the spectrum of Δ there is discrete. Hence some eigenvalue λ_1 satisfies $|\lambda_1 - \frac{1}{4}| \leq \|(\Delta - \frac{1}{4})F_S\|/\|F_S\|$; now use Lemmas 5 and 7. All cusp forms of prime level with a primitive character of that conductor are new, and the λ_1 -eigenspace is finite dimensional and Hecke stable. $\square$

Remark 8. Compared with [1]: (i) [1] already notes $C_{f,1} = 1$ for odd forms on $\operatorname{PSL}_2(\mathbb{Z})$. Here that is decisive. For even forms with this character the Eisenstein families $E_{1,\chi}$ and $E_{\chi,1}$ share a Hecke eigenvalue only at primes with $\chi(\ell) = 1$, the smallest being $\ell = 31$, and the threshold $\ell e^\delta/\sqrt{N}$ of the lower bound would then sit in the exponentially small tail of K_0 . (ii) Their Lemma 3 bounds $\|(\Delta - \lambda)f_S\|$ by the supremum of the defect over a two-dimensional δ -neighbourhood of the boundary. Lemma 4 replaces it with L^2 norms on the sides themselves. That avoids evaluating $\tilde{f}$ below the domain, where a truncated Hejhal solution is unreliable. It also avoids tracking which tile contains each point near the 299 vertex cycles. (iii) The constraint $\delta \leq 1/(4\sqrt{\lambda})$ of [1] is not needed, because $\hat{k}$ is bounded below directly.

6 Numbers

With the $M = 8000$ float64 coefficients (antisymmetrised), $\ell_{\max} \leq n\pi \sinh \delta$ where n is the number of tiles meeting one ball of radius δ ($n = 6$ observed for $\delta \leq 0.1$, 10–11 for $0.2 \leq \delta \leq 0.5$; a rigorous count is part of the certification), the theorem reads $|\lambda_1 - \frac{1}{4}| \leq c_0\sqrt{Q_0} + c_1\sqrt{Q_1}$ with:

δ	n	k_0	α_0	α_1	c_0	c_1
0.05	6	7.85×10^{-3}	6.62	0.149	3.55×10^2	7.99
0.1	6	3.14×10^{-2}	9.37	0.422	1.31×10^2	5.91
0.2	10	1.25×10^{-1}	17.17	1.547	6.58×10^1	5.93
0.3	12	2.82×10^{-1}	23.19	3.132	4.34×10^1	5.86
0.5	12	7.77×10^{-1}	30.57	6.875	2.51×10^1	5.64

Since $\sqrt{Q_0} = \sqrt{L}\eta_{\text{rms}}$ with $\sqrt{L} = 35.92$, a root-mean-square jump of 10^{-12} with a normal-derivative jump of 4×10^{-11} would give $|\lambda_1 - \frac{1}{4}| \leq 9 \times 10^{-9}$ at $\delta = 0.5$. The Fricke arc is one of the 921 sides. On its top quarter ($\pi/4 \leq \arg z \leq 3\pi/4$, length 1.76 of 8.99) the jump has been bounded rigorously: 1.571×10^{-6} for the float64 coefficients. With the double-double coefficients the jump at the 60,000 box centres of the same arc is at most 1.9×10^{-12} ($M = 8000$) and 7.5×10^{-13} ($M = 12000$).

7 What remains, and what this does not prove

Remaining for a complete certificate. (a) Rigorous upper bounds for Q_0 and Q_1 : on each side, D_s and $\partial_\nu D_s$ are evaluated at Chebyshev points by cancelling sums with bounded float64 error. They are interpolated, and the interpolation error is bounded through a crude bound on a Bernstein ellipse, where $|K_0(w)| \leq K_0(\text{Re } w)$ and $|e(nx)| \leq e^{2\pi|n||\text{Im } x|}$. About 70 nodes per unit length suffice: roughly 10^5 nodes over $L = 1290.55$, each needing $\tilde{f}$ and $\nabla \tilde{f}$ at two points of the floor. (b) A rigorous bound for $\ell_{\max}(\delta)$ by enumerating the tiles within δ of $\mathfrak{F}^+$. (c) The same with the double-double coefficients.

What the theorem gives. Existence of an odd Maass cusp newform on $\Gamma_0(1951)$ with nebentypus χ_{780} whose eigenvalue lies in an explicit interval around $1/4$. It does not show that the eigenvalue equals $1/4$, that the form's Hecke eigenvalues are the traces of the icosahedral representation, or anything about Artin's conjecture for it: an enclosure cannot certify an equality, and $1/4$ lies inside every interval this method can produce.

References

- [1] A. R. Booker, A. Strömbergsson, A. Venkatesh, *Effective computation of Maass cusp forms*, Int. Math. Res. Not. 2006, Art. ID 71281.